Science Notebook

ChatGPT edition

Evidence before energy

SUPERVISED-ONLY PROJECT

One named, sober adult controls the launcher, fuel, ignition, commands, misfires, and cease-fire. Minors do not fuel, ignite, clear a misfire, or handle a loaded launcher. This set is educational guidance, not a certification of any component or assembly.

THE LEARNING SEQUENCE

Complete the paper models and cold measurements first. Live operation is optional, adult-controlled, outdoors only, and permitted only after every gate on the **Range Command Card** passes. Never change fuel, projectile fit, angle, or hardware merely to chase a larger result.

Model 1 · A bounded flame

Combustion is possible only inside a fuel's flammable range. Below its lower limit the mixture is too lean; above its upper limit it is too rich. That makes "more fuel" scientifically wrong as a general rule. Product formulation, sprayer, temperature, chamber history and mixing all vary, so this edition provides no universal dose.

Too lean · bounded region · too rich. The drawing is conceptual, not a dose scale.

Write before observing

Claim	A bounded mixture can ignite; mixtures outside the bounds may not.
Prediction	Adding fuel can move a mixture toward *or away from* the flammable region.
Safety consequence	Never top up a misfire; hold 60 seconds, vent fully, and return to the cold state. Never use liquid accelerants or oxygen enrichment.

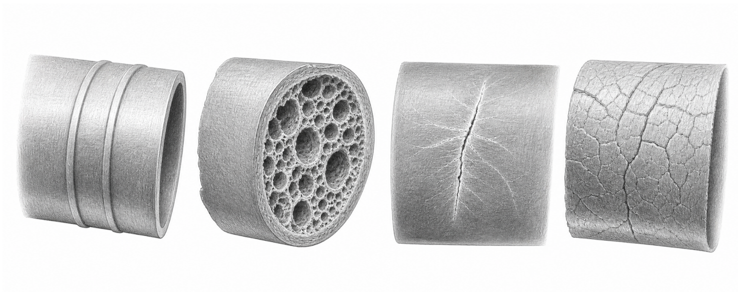
Model 1 record

Draw a number line with an unnamed bounded region. Mark an initial mixture and two possible moves caused by "adding more." Why can the same action move toward safety in one sketch and away in the other?

What observation would *not* justify adding fuel? ______________________________

Model 2 · Pressure is not approval

Heating gas in a constrained volume can raise pressure, but a plumbing pressure mark describes a listed plumbing product under specified conditions. It does not certify a homemade combustion assembly. Use only solid-wall, pressure-marked pipe *and* pressure fittings; never DWV/cellular core and never compressed air or gas. Temperature, sunlight, impact, workmanship and chemical exposure can reduce margin.



Sound surface · cellular core · stress crack · crazing. A mark cannot cancel a defect.

Material audit

Evidence read directly from every part	Pass/fail	Adult initials
Solid wall; pressure standard/rating; matching pressure fitting		
No DWV, cellular/foam core, cracks, whitening, crazing or deformation		
Cement/primer compatible; batch usable; label instructions retained		
Cure start recorded; 48 h minimum and manufacturer longer time satisfied		

Model 2 inspection questions

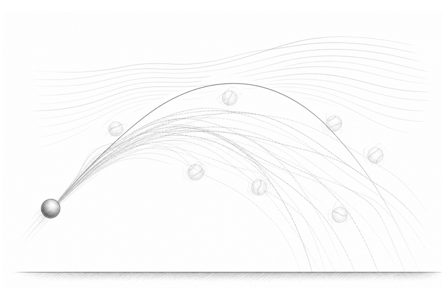
Choose one real component and copy, do not paraphrase, its pressure marking: ____________
Under raking light, sketch any molding line and distinguish it from a crack. What changed when the part was rotated?
Can a successful prior use prove the absence of accumulated damage? Explain: ____________

Model 3 · A useful trajectory, with a boundary

For launch speed v, angle θ, and gravity g, the no-drag model is

$$x(t) = v\cos(\theta)t, \qquad y(t) = v\sin(\theta)t - \tfrac{1}{2}gt^2.$$

It explains the parabolic ideal and separates horizontal from vertical motion. It does not include drag, potato shape, wind, launch-height difference, tumble, fragmentation, or shot-to-shot variation.



One ideal path versus a family of plausible outcomes under omitted effects.

Evidence table — observations, not performance tuning

Trial	Cold/live	Predicted observation	Observed / uncertainty
1			
2			
3			

Model 3 uncertainty record

Omitted variable	Direction predictable?	How the range gate handles it
Drag / irregular shape		
Wind / tumble		
Fragmentation / bounce		

Intermediate question: if three observations cluster, does that establish a safe boundary? Why not? ____________

Model 4 · Controls beat measurement

Impulse sound can injure immediately, and ordinary meters may clip the peak. Therefore a phone reading cannot clear the hazard. Distance, limited attendance, earplugs plus earmuffs, impact-rated eye protection, and a disciplined firing line remain required regardless of the displayed number.

Layered controls remain necessary when a phone waveform clips or misses the peak.

Model 4 measurement limits	
Displayed phone peak	______
Sampling / clipping clue	______
What remains unknown	______
Controls retained regardless	distance; limited attendance; impact-rated eye protection; earplugs plus earmuffs; firing line

A good conclusion

State what the model explained, what the evidence showed, and what remains unknown. The strongest result is often a justified stop decision. Do not report a PVC pressure rating as a combustion rating, a computed range as containment, or one successful firing as proof of safety.

Final evidence acceptance

For each model I recorded a prediction before observation, separated direct evidence from inference, named at least one uncertainty, and retained every safety control when the measurement could not resolve the hazard.

Most important stop decision: ______

Student / recorder ______ Adult review ______ Date ______